

[GETNET] APLICANDO PSU 19.28 EM LINUX

△OBS1:

RAC

Nó 01 aplica psu com user root

Nó 02 aplica psu, neste ponto ele aplica o datapatch pelos 2

Nó 01, aplica OJVM

Nó 02, aplica OJVM

Nó 02, roda o datapatch para o OJVM

△OBS2:

cat /etc/redhat-release

em rac se for maior ou igual 8.8

su - grid

. grid_profile

export CV_ASSUME_DISTID=OEL8.1

echo \$CV_ASSUME_DISTID

0) PRIMEIRAS VALIDAÇÕES PRE PSU

0.1) Identificar o RHEL do Host

```
cat /etc/redhat-release
```

0.2) Validar se o banco esta up e identificar as características

```
su - oracle
```

```
./home/oracle/db_profile
```

```
ps -ef | grep pmon
```

```
srvctl config database
```

```
srvctl config database -d
```

```
sqlplus -v
```

Validar se tem Wallet no Host:

```
sqlplus / as sysdba
SET LINES 800
SET PAGESIZE 10000
BREAK ON REPORT
COMPUTE SUM LABEL TOTAL OF GAP ON REPORT
select primary.thread#,
primary.maxsequence primaryseq,
standby.maxsequence standbyseq,
primary.maxsequence - standby.maxsequence gap
from ( select thread#, max(sequence#) maxsequence
from v$archived_log
where archived = 'YES'
and resetlogs_change# = ( select d.resetlogs_change# from v$database d )
group by thread# order by thread# ) primary,
( select thread#, max(sequence#) maxsequence
from v$archived_log
where applied = 'YES'
and resetlogs_change# = ( select d.resetlogs_change# from v$database d )
group by thread# order by thread# ) standby
where primary.thread# = standby.thread#;
set lines 300
SELECT DATABASE_ROLE, DB_UNIQUE_NAME INSTANCE, OPEN_MODE, PROTECTION_MODE, PROTECTION_LEVEL, SWITCHOVER_S
TATUS FROM V$DATABASE;
SELECT status FROM V$ENCRYPTION_WALLET;
```

0.3) Verificar se precisa de espaço para a aplicação

```
df -h
```

0.4) Verificar possíveis contingências

```
cat $ORACLE_HOME/network/admin/tnsnames.ora
```

0.5) Validar o status do GRID

```
su - grid

./home/grid/grid_profile

crsctl status res -t

ps -ef | grep grid
```

0.6) Validação do Compatible

```
set lines 300
col name format a30
col value format a20
col description format a100
SELECT name, value, description FROM v$parameter WHERE name = 'compatible';
```

NAME	VALUE	DESCRIPTION
compatible	12.1.0.2.0	Database will be completely compatible with this software version

```
sqlplus / as sysasm
```

```
col VALUE format a30;
col COMPATIBILITY form a10
col DATABASE_COMPATIBILITY form a10
col NAME form a20
```

```
select group_number, name, compatibility, database_compatibility from v$asm_diskgroup;
```

```
select g.name,a.name,a.value from v$asm_diskgroup g, v$asm_attribute a where g.group_number=a.group_number and a.name like '%compat%';
```

1) LIMPEZA DA PASTA DE PATCHES:

```
su - grid

. grid_profile

cd /u02/oracle/patches

pwd

ll -la

rm -rf /u02/oracle/patches/*
```

2) COPIA DOS ARQUIVOS

Linux

- Opatch (v47)

wget https://repository.getnet.com.br/binarios/oracle/odb/PST19C_TEMPLATE/p6880880_190000_Linux-x86-64.zip
- Patch Grid (19.28)
wget https://repository.getnet.com.br/binarios/oracle/odb/PST19C_TEMPLATE/p37957391_190000_Linux-x86-64.zip
- Patch OJVM (19.28)
wget https://repository.getnet.com.br/binarios/oracle/odb/PST19C_TEMPLATE/p37847857_190000_Linux-x86-64.zip
- Patch Java JDK (19.28)
wget https://repository.getnet.com.br/binarios/oracle/odb/PST19C_TEMPLATE/p37860476_190000_Linux-x86-64.zip
- PSU GGS (19.27)
wget https://repository.getnet.com.br/binarios/oracle/odb/PST19C_TEMPLATE/p37777884_1927000OGGRU_Linux-x86-64.zip

Linux

wget https://repository.getnet.com.br/binarios/oracle/odb/PST19C_TEMPLATE/p6880880_190000_Linux-x86-64.zip
wget https://repository.getnet.com.br/binarios/oracle/odb/PST19C_TEMPLATE/p37957391_190000_Linux-x86-64.zip
wget https://repository.getnet.com.br/binarios/oracle/odb/PST19C_TEMPLATE/p37847857_190000_Linux-x86-64.zip
wget https://repository.getnet.com.br/binarios/oracle/odb/PST19C_TEMPLATE/p37860476_190000_Linux-x86-64.zip
wget https://repository.getnet.com.br/binarios/oracle/odb/PST19C_TEMPLATE/p37777884_1927000OGGRU_Linux-x86-64.zip

▲ AHF Julho/2023

```
scp adm.as58172@GNCASHWL01962:/u02/ORAEX/db/19c/Linux/PSU/AHF-LINUX_v23.9.0.zip /u02/oracle/patches/AHF-LINUX_v23.9.0.zip
```

2.1) unzip dos arquivos

Linux

```
unzip /u02/oracle/patches/p37957391_190000_Linux-x86-64.zip
unzip /u02/oracle/patches/p37847857_190000_Linux-x86-64.zip
unzip /u02/oracle/patches/p37860476_190000_Linux-x86-64.zip
y
```

3) UNZIP – USER ROOT:

OPATCH 39

Profile do Banco

```
su - root
./home/oracle/db_profile
cd $ORACLE_HOME
pwd
/u01/app/oracle/product/19.3.0/dbhome_1
$ORACLE_HOME/OPatch/patch version
rm -rf OPatch
rm -rf OPatch.old
rm -rf OPatch__old
rm -rf p6880880_190000_Linux-x86-64.zip
cp /u02/oracle/patches/p6880880_190000_Linux-x86-64.zip .
unzip p6880880_190000_Linux-x86-64.zip
chown -R oracle:oinstall OPatch
rm -rf p6880880_190000_Linux-x86-64.zip
$ORACLE_HOME/OPatch/patch version
```

Profile do GRID

```

su - root
./home/grid/grid_profile
cd $ORACLE_HOME
pwd
/u01/app/19.16.0/grid
$ORACLE_HOME/OPatch/opatch version
rm -rf OPatch
rm -rf OPatch.old
rm -rf OPatch__old
rm -rf p6880880_190000_Linux-x86-64.zip
cp /u02/oracle/patches/p6880880_190000_Linux-x86-64.zip .
unzip p6880880_190000_Linux-x86-64.zip
chown -R grid:oinstall OPatch
rm -rf p6880880_190000_Linux-x86-64.zip
$ORACLE_HOME/OPatch/opatch version

```

4) BAIXAR GGS E BANCO PARA O SNAPSHOT

GGs

```

su - oracle

./db_profile

cd /ggs

./ggsci

stop *

stop manager

```

(caso necessário, kill -9 nos processos resultantes do: ps -ef |grep ggs)

Banco

```

su - oracle

./db_profile

ps -ef | grep pmon

sqlplus / as sysdba

shutdown immediate;

```

Exemplos de outras maneiras:

```

history | grep stop
history | grep srvctl

srvctl stop instance -d HCTPRD -node gncasspa03482
srvctl stop instance -d HCTPRD -i HCTPRD1

```

*****TIRAR SNAPSHOT NO TS 38*****

△Dica:
 Limpeza nos patches inativos.
 Como usuário Oracle:
 - Listar os patches antigos e inativos:

```
$ORACLE_HOME/OPatch/opatch util listorderedinactivepatches
```

- Deletar os patches antigos e inativos:

5) APLICAR O PATCH COM ROOT

```
su - root
./home/grid/grid_profile
```

5.1) Correção das permissões do GRID

```
cd $ORACLE_HOME
cd ..
ll
chown grid:oinstall grid
chmod 775 grid
cd $ORACLE_HOME/OPatch
```

PSU

```
./opatchauto apply /u02/oracle/patches/37957391
```

JDK

```
./opatchauto apply /u02/oracle/patches/37860476
```

Rerun se tiver dado erro:

```
cd $ORACLE_HOME/OPatch
./opatchauto resume
```

6) APLICAR O OJVM COM ORACLE

6.1) Baixar o banco oracle e aplicar

1. `su - oracle`
2. `./home/oracle/db_profile`
3. `ps -ef | grep pmon`
4. `sqlplus / as sysdba`
5. `shutdown immediate;`

OJVM

1. `cd $ORACLE_HOME/OPatch`
2. `./opatch apply /u02/oracle/patches/37847857`

Caso seja Standby:

1. `sqlplus / as sysdba`
2. `startup nomount`
3. `alter database mount standby database;`
4. `alter database open read only;`
5. `ALTER DATABASE RECOVER MANAGED STANDBY DATABASE DISCONNECT FROM SESSION;`
6. `SELECT PROCESS, STATUS, SEQUENCE# FROM V$MANAGED_STANDBY;`

Caso seja Primary:

1. `su - oracle`
2. `./home/oracle/db_profile`
3. `ps -ef | grep pmon`
4. `sqlplus / as sysdba`
5. `startup;`

OBS: Caso tenha Wallet no host, sera necessário abri-lo após o banco tiver UP

Subir o GGS

1. `su - oracle`
2. `./db_profile`
3. `cd /ggs`
4. `./ggsci`
5. `start manager`
6. `start *`

Ao final, exclusivamente no Primary, rodar o Datapatch

1. `su - oracle`
2. `./home/oracle/db_profile`
3. `cd $ORACLE_HOME/OPatch`
4. `./datapatch -verbose`

COLETA DE EVIDENCIAS:

1. `su - oracle`
2. `./home/oracle/db_profile`

```

3. sqlplus / as sysdba
4. col HOST_NAME format a30
5. set lines 230
6. col COMP_NAME format a55
7. set pages 900
8. col DESCRIPTION format a65
9. col ACTION_TIME format a29
10. col COMMENTS format a40
11. col ACTION format a9
12. col NAMESPACE format a9
13. col ID format 99999999
14. select instance_name, host_name, version, status, open_mode, database_role, switchover_status from gv$instance, v$database;
15. select comp_name, version, status, modified from dba_registry;
16. select PATCH_ID, ACTION, STATUS, ACTION_TIME, DESCRIPTION from dba_registry_sqlpatch;

```

Retorno:

```

INSTANCE_NAME HOST_NAME VERSION STATUS OPEN_MODE DATABASE_ROLE SWITCHOVER_STATUS
-----
MINHACTAUDIT GNCASSPL04107 19.0.0.0.0 OPEN READ WRITE PRIMARY TO STANDBY
SQL> SQL>
COMP_NAME VERSION STATUS MODIFIED
-----
Oracle Database Catalog Views 19.0.0.0.0 VALID 25-MAR-2020 14:05:25
Oracle Database Packages and Types 19.0.0.0.0 VALID 25-MAR-2020 14:05:25
Oracle Real Application Clusters 19.0.0.0.0 OPTION OFF 17-APR-2019 02:03:49
JServer JAVA Virtual Machine 19.0.0.0.0 VALID 31-OCT-2023 02:39:25
Oracle XDK 19.0.0.0.0 VALID 25-MAR-2020 14:05:26
Oracle Database Java Packages 19.0.0.0.0 VALID 25-MAR-2020 14:05:26
OLAP Analytic Workspace 19.0.0.0.0 VALID 25-MAR-2020 14:05:27
Oracle XML Database 19.0.0.0.0 VALID 25-MAR-2020 14:05:26
Oracle Workspace Manager 19.0.0.0.0 VALID 25-MAR-2020 14:05:26
Oracle Text 19.0.0.0.0 VALID 25-MAR-2020 14:05:26
Spatial 19.0.0.0.0 VALID 25-MAR-2020 14:05:27
Oracle OLAP API 19.0.0.0.0 VALID 16-JUL-2022 00:52:07
Oracle Label Security 19.0.0.0.0 VALID 25-MAR-2020 14:05:27
Oracle Database Vault 19.0.0.0.0 VALID 25-MAR-2020 14:05:27
14 rows selected.
SQL> SQL>
PATCH_ID ACTION STATUS ACTION_TIME DESCRIPTION
-----
29517242 APPLY SUCCESS 25-MAR-20 01.26.18.459047 PM Database Release Update : 19.3.0.0.190416 (29517242)
30557433 APPLY SUCCESS 25-MAR-20 02.05.16.720198 PM Database Release Update : 19.6.0.0.200114 (30557433)
33806152 APPLY SUCCESS 16-JUL-22 12.53.12.500615 AM Database Release Update : 19.15.0.0.220419 (33806152)
34086870 APPLY SUCCESS 25-OCT-22 01.43.07.158979 AM OJVM RELEASE UPDATE: 19.16.0.0.220719 (34086870)
34133642 APPLY SUCCESS 25-OCT-22 01.43.07.155018 AM Database Release Update : 19.16.0.0.220719 (34133642)
34765931 APPLY SUCCESS 12-APR-23 01.38.55.591966 AM DATABASE RELEASE UPDATE : 19.18.0.0.230117 (REL-JAN230131) (34765931)
34086870 ROLLBACK SUCCESS 12-APR-23 01.54.37.206416 AM OJVM RELEASE UPDATE: 19.16.0.0.220719 (34086870)
34786990 APPLY SUCCESS 12-APR-23 01.54.37.210340 AM OJVM RELEASE UPDATE: 19.18.0.0.230117 (34786990)
35042068 APPLY SUCCESS 07-JUN-23 01.58.11.158054 AM Database Release Update : 19.19.0.0.230418 (35042068)
34786990 ROLLBACK SUCCESS 07-JUN-23 03.16.10.429853 AM OJVM RELEASE UPDATE: 19.18.0.0.230117 (34786990)
35050341 APPLY SUCCESS 07-JUN-23 03.16.10.433823 AM OJVM RELEASE UPDATE: 19.19.0.0.230418 (35050341)
35050341 ROLLBACK SUCCESS 12-AUG-23 02.18.17.072677 AM OJVM RELEASE UPDATE: 19.19.0.0.230418 (35050341)
35354406 APPLY SUCCESS 12-AUG-23 02.18.19.369834 AM OJVM RELEASE UPDATE: 19.20.0.0.230718 (35354406)
35320081 APPLY SUCCESS 12-AUG-23 02.18.19.365955 AM Database Release Update : 19.20.0.0.230718 (35320081)
35354406 ROLLBACK SUCCESS 31-OCT-23 02.40.46.615205 AM OJVM RELEASE UPDATE: 19.20.0.0.230718 (35354406)
35648110 APPLY SUCCESS 31-OCT-23 02.40.47.490862 AM OJVM RELEASE UPDATE: 19.21.0.0.231017 (35648110)
35643107 APPLY SUCCESS 31-OCT-23 02.40.47.486601 AM Database Release Update : 19.21.0.0.231017 (35643107)
17 rows selected.

```

OBS: Se na evidencia o Spatial estiver como LOADING, rodar isso aqui:

```

1. #### Spatial
2. set serveroutput on
3. exec sys.VALIDATE_SDO();

```

7) EXECUTAR OS PASSOS DE COMPACTAÇÃO DO PATCHSTORAGE

```

1. su - root
2. ./home/grid/grid_profile
3. cd $ORACLE_HOME/patch_storage
4. pwd
5. ls -la
6. tar -zcvf patch_storage.tar.gz .

```

```

7. ls -la
8. rm -rf 2*
9. rm -rf 3*
10. su - root
11. ./home/oracle/db_profile
12. cd $ORACLE_HOME/patch_storage
13. pwd
14. ls -la
15. tar -zcvf patch_storage.tar.gz .
16. ls -la
17. rm -rf 2*
18. rm -rf 3*

```

8) LIMPEZA DA PASTA DE PATCHES:

```

1. su - grid
2. ./grid_profile
3. cd /u02/oracle/patches
4. pwd
5. ll -la
6. rm -rf /u02/oracle/patches/*

```

```

1. su -root
2. cd /u02/tmp/
3. ll
4. rm -rf /u02/tmp/*

```

```

1. su - root
2. cd /home/oracle/
3. ll
4. rm -rf /home/oracle/old_db_profile

```

```

1. cd /home/grid/
2. ll
3. rm -rf /home/grid/old_grid_profile

```

GGs

```

su - oracle
bash
. db_profile
cd /u02/oracle/patches/
ls -la
rm -rf /u02/oracle/patches/*

```

```

J45bAEyvWOcpKD*
#### Cópia dos binários
- Opatch (v46)
wget https://repository.getnet.com.br/binarios/oracle/odbc/PST19C_TEMPLATE/p6880880_190000_Linux-x86-64.zip

```

```

Patch GGS 19.27
wget https://repository.getnet.com.br/binarios/oracle/odbc/PST19C_TEMPLATE/p37777884_1927000OGGRU_Linux-x86-64.zip

```

```

#### Unzip do binário
unzip /u02/oracle/patches/p37777884_1927000OGGRU_Linux-x86-64.zip
#### Opatch /ggs
cd /ggs
/ggs/OPatch/patch version
rm -rf OPatch
rm -rf p6880880_190000_Linux-x86-64.zip
cp /u02/oracle/patches/p6880880_190000_Linux-x86-64.zip .
unzip p6880880_190000_Linux-x86-64.zip
rm -rf p6880880_190000_Linux-x86-64.zip
/ggs/OPatch/patch version

```

```

#### Stop Golden Gate
cd /ggs
./ggsci
info all
stop *
stop manager

```

```

|

```

```

ps -ef |grep ggs

```

```

#### Ajuste Opatch (atualmente na versão 37)

```

```

export ORACLE_HOME=/ggs
export PATH=$ORACLE_HOME/OPatch:$PATH
which opatch

```

```

"/ggs/OPatch/patch"

```

/ggs/OPatch/patch version
/ggs_acfs/OPatch/patch version

or

opatch version -jdk \$ORACLE_HOME/jdk

"OPatch Version: 12.2.0.1.39"

```
cd /u02/oracle/patches/37777884
opatch prereq CheckConflictAgainstOHWithDetail -ph ./
```

"Oracle Interim Patch Installer version 12.2.0.1.39

Copyright (c) 2023, Oracle Corporation. All rights reserved.

PREREQ session

Oracle Home : /ggs

Central Inventory : /u01/app/oraInventory

from : /ggs/oraInst.loc

OPatch version : 12.2.0.1.39

OUI version : 12.2.0.4.0

Log file location : /ggs/cfgtoollogs/patch/opatch2023-05-18_12-03-37PM_1.log

Invoking prereq "checkconflictagainsthwithdetail"

Prereq "checkConflictAgainstOHWithDetail" passed.

OPatch succeeded."

```
opatch apply
```

"Oracle Interim Patch Installer version 12.2.0.1.39

Copyright (c) 2023, Oracle Corporation. All rights reserved.

Oracle Home : /ggs

Central Inventory : /u01/app/oraInventory

from : /ggs/oraInst.loc

OPatch version : 12.2.0.1.39

OUI version : 12.2.0.4.0

Log file location : /ggs/cfgtoollogs/patch/opatch2023-05-18_12-06-12PM_1.log

Verifying environment and performing prerequisite checks...

OPatch continues with these patches: 34962407

Do you want to proceed? [y/n]

y

User Responded with: Y

All checks passed.

Please shutdown Oracle instances running out of this ORACLE_HOME on the local system.

(Oracle Home = '/ggs')

Is the local system ready for patching? [y/n]

y

User Responded with: Y

Backing up files...

Applying interim patch '34962407' to OH '/ggs'

Patching component oracle.oggc core.ora19c, 19.1.0.0.0...

Patch 34962407 successfully applied.

Sub-set patch [34377379] has become inactive due to the application of a super-set patch [34962407].

Please refer to Doc ID 2161861.1 for any possible further required actions.

Log file location: /ggs/cfgtoollogs/patch/opatch2023-05-18_12-06-12PM_1.log

OPatch succeeded."

Start Golden Gate

```
su - oracle
bash
. db_profile
cd /ggs
./ggsci
info all
start manager
start *
```

```
su - root
. /home/oracle/db_profile
cd /u02/oracle/patches/
ls -la
rm -rf /u02/oracle/patches/*
ls -la
```

Evidência do host desatualizado na HOME do Golden Gate :

```
/ggs/jdk/jre/bin/java -version
java version "1.8.0_441"
Java(TM) SE Runtime Environment (build 1.8.0_441-b07)
Java HotSpot(TM) 64-Bit Server VM (build 25.441-b07, mixed mode)
```

Evidência do host atualizado na HOME do Oracle Database :

```
$ORACLE_HOME/jdk/jre/bin/java -version
java version "1.8.0_451"
Java(TM) SE Runtime Environment (build 1.8.0_451-b10)
Java HotSpot(TM) 64-Bit Server VM (build 25.451-b10, mixed mode)
*** stop dos processos do Golden Gate, stop do manager e kill dos processos ggs.log
```

```
cd /ggs

mv jdk_jdk_old
cp -r /u01/app/oracle/product/19.16.0/dbhome_1/jdk .
tar -zcvf /u02/oracle/patches/jdk_ggs.tar.gz jdk_old
ls -la /u02/oracle/patches/
```

```
rm -rf jdk_old
/ggs/jdk/jre/bin/java -version
```